

I-CovidVis – A Visual Analytics Tool for Interoperable Healthcare Databases using Graphs

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Abstract—The current COVID-19 pandemic has promoted the periodic release of several health databases aimed at discovering relationships in the data, detecting similar problems in patients, and studying the evolution of the disease. A way to exploit the data is to use visualization techniques, which can lead to the discovery of insights and patterns, as well as to guide analysis procedures to understand the data. In this paper, we present I-CovidVis, a visualization tool to explore data from interoperable healthcare systems, able to compare and navigate in both global and local perspectives. Our approach is to model data as a graph and explore its structural and temporal views. Our proposal facilitates the perception of patterns, trends, periodicity, and anomalies, resulting in faster decision making.

Index Terms—information visualization, visual analytics, data visualization, complex networks, graph visualization, healthcare systems, data mining.

I. INTRODUCTION

The increasing growth of healthcare databases motivated by the COVID-19 pandemic demands the integration and manipulation of data made available by several institutions around the world, to support data interoperability and comprehension [1]. There are tools for data standardization, such as those based on the Observational Medical Outcomes Partnership Common Data Model (OMOP-CDM) [2]. However, these approaches lack visual aids to assist users such as doctors, hospital managers, and others in analyzing data.

Healthcare data demand complex structures, so modeling them as graphs may help to reduce their complexity and to increase the user's understanding of many aspects [3]. Graphs use nodes to indicate the instances of the modeled objects, and edges to represent the relationships between them. Complex networks containing information about *when* the relationships occur are called temporal networks. Graph analyses take advantage of several statistical measures and techniques to improve analytical processes, such as community detection [4].

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The communities in the graphs represent groups of instances that connect more frequently with each other than with other groups.

Information Visualization can also play a crucial role in understanding, empowering the users to visualize and navigate the data through interactive techniques, improving understanding, reasoning, and decision making [5]. Recent studies have shown the importance of visual analysis for health systems and several tools have been proposed to explore complex data sets [6]. However, most recent systems are focused on simulating the infection spread and studying the disease behavior through statistical modeling or clustering analyses [6], [7].

Graph Visualization is well-suited to understand flexible data structures and to handle interoperable systems. Graph-based techniques and tools are widely used for exploratory data analysis, to find patterns, trends, and anomalies in the data [8]. Several techniques were proposed and improved over the years, such as node/edge organization and sampling [4], and several tools support visualization and interaction, such as Gephi, Cytoscape, and DyNetVis [9]. Graph visualization has been used in healthcare systems, for example, to study infection propagation patterns during international flights [10].

In this paper, we introduce *Interoperable Covid Visualizer (I-CovidVis)*, a novel tool for exploratory visual analyses on interoperable healthcare databases, able of handling different exams and measurement targets, using a graph structure to model the data. Although its concept can be applied to analyze other types of data, I-CovidVis was designed with a focus on the need to share information and research efforts required by the remarkable urgency of understanding the covid pandemic. We also evaluated a case study based on the FAPESP COVID-19 Data Sharing/BR repository [11], [12], containing data from three healthcare institutions in Brazil, with almost 10 million clinical exams from patients related to COVID-19.

I-CovidVis includes resources to clean and sampling data, important preliminary steps for visual analysis. It represents the data as a graph, allowing to identify relationships to sim-

plify the analysis. Finally, it generates insightful visualizations based on graph visualization techniques, allowing the analysts to interact and coordinate analysis among different visual representations. I-CovidVis aims at exploring the relationships between patients and exams, allowing detecting visual patterns from global to local perspectives.

The paper is organized as follows: Section II describes related works and concepts of this paper. Section III discusses I-CovidVis and the visualization techniques employed. Section IV presents the data mining operations involved, exemplifies the visual analyses, and shows results for our case study. Section V provides some discussions and limitations. Section VI concludes the paper.

II. RELATED WORK

One of the most common approaches to study Covid-19 is estimating the pandemic risks by calculating infection model propagation, probability, and so on [13]. Visualization information techniques have been used to alert the world population about potential risks from covid, creating illustrations using bar charts, interactive and animated maps, and others [14].

Complex networks are often employed to model the relationships among instances. In the context of covid, studies use graphs to model the countries' airports to assess flight security among countries, relating the number of confirmed cases and density in each country [15]. Some studies extend those analyses to visualization perspectives, showing maps with the correlation of changes in the statistics of confirmed covid cases among neighbor countries [10].

The *FAPESP COVID-19 Data Sharing/BR* database [11] contains deidentified patient information related to covid, such as clinical exams, personal and demographic information (age, sex, birthplace), from three large Brazilian healthcare institutions [12]. Studies that used this data set have shown some data representation problems, highlighting the importance of a data cleaning for prospective analyses. There is also a study using this database relating sex, age, and immunological response of patients with COVID-19 [16].

III. GRAPH VISUALIZATION TECHNIQUES

The commonest approach to visualize graphs is the node-link diagram (also called structural layout), where nodes are represented as circles and the edges as straight lines. This representation is good to visualize the network structure and the relationships among objects (Fig. 1(b)). One of the most common techniques to enhance the quality of node-links is the force-directed layout, which uses the idea of springs separating the nodes in the layout, using repulsive and attractive forces among nodes [17]. This technique approximates related nodes, spotting groups (similar to community detection algorithms).

Timeline representations are widely used to understand the temporal evolution of graphs [18]. The Massive Sequence View (MSV) layout (also called temporal layout) is very good to represent evolution over time, where the nodes are positioned vertically and timestamps are positioned horizontally (Fig. 1(c)) [9]. An important strategy for MSV layouts

requires node order, aiming at reducing the edge overlapping to improve readability, which is ideal for large graphs [4], [9].

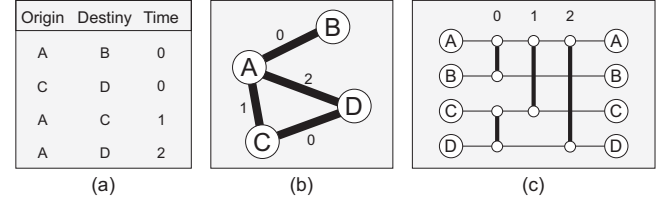


Fig. 1. Techniques for graph visualization: (a) Tabular representation; (b) node-link diagram; (c) Massive Sequence View (MSV) layout.

IV. I-COVIDVIS APPLIED TO A CASE STUDY

Here we describe I-CovidVis, showing its usefulness to analyze data obtained from [11]. It is based on an extension/specialization of the Dynamic Network Visualization (DyNetVis) software [9] for interoperable data about the COVID-19 disease. The targets for I-CovidVis are physicians, hospital managers, administrative staff, data analysts, and researchers.

A. Original data set

In its current release (August 2020), the FAPESP COVID-19 Data Sharing/BR database contains data from three healthcare institutions: Albert Einstein Hospital, Fleury laboratories, and Sirio-Libanês Hospital, called in this work as HF1, HF2, and HF3, respectively. Together, there are data from more than 332 thousand patients and almost 10 million laboratory results. Each laboratory result has its exam identifier, the evaluated analyte, the measured and reference values, and the unit. One of the institutions (HF3) also provided the patient outcome. Table I shows the cardinality of each entity type, and the average and the median amount of entities per patient.

TABLE I
THE NUMBER OF INSTANCE TYPES, AND THE AVERAGE NUMBER OF RESULTS PER PATIENT IN THE THREE HEALTHCARE INSTITUTIONS. HF1 AND HF2 DO NOT INCLUDE PATIENT OUTCOMES.

Entity	HF1	HF2	HF3
patients	79,863	247,981	4,273
exams	63	763	537
analytes	126	1,031	731
results	3,415,155	5,873,403	631,352
outcomes	-	-	16,957
exams/patient (median)	4.9 (1)	9.1 (2)	14.4 (5)
analytes/patient (median)	15.4 (1)	18.4 (4)	32.8 (24)
results/patient (median)	42.8 (2)	23.7 (4)	149.7 (24)
outcomes/patient (median)	-	-	4.0 (3)

B. Data cleaning and normalization

Each healthcare institution uses different types of exams, analytes (often with diverse vocabularies and spellings), and units (mostly SI but with different scales). The labels also contain noise, such as character typos, extraneous spaces, and punctuation. To analyze and compare the patients' conditions, we used the I-CovidVis resources to clean the values and

normalize comparable laboratory results, unifying the representation and scales.

After the analyte selection and normalization, we managed to reduce the noisy exam-analyte source (Table I) from 2,643 to the clean 73 standardized analytes described in the literature. The cleaned data include approximately 2.5 million laboratory results from the original 10 million. By considering an indicator matrix for how many patients any given analyte was recorded (analyte $\times$ patient), we identified the analytes with records for at least 0.1% of patients (332 patients, a small but significant sample).

For network visualization purposes, the FAPESP COVID-19 data set is very large. Thus we used a guided sampling [20] from the laboratory tests, preserving the proportion of tests performed at each health care institution. Our goal here is to ensure both that most tests are present in the sample and also to get an overview of the data. We limited the sampling to 1% of the data set to not overload the visualization processes, such as the node organization algorithm (force-directed layout), whereas avoiding visual cluttering.

The goal of our analysis was to understand the relationship between patients and exams in different healthcare institutions, so the data were modeled as a single directed graph, where each edge represents an exam result performed at a given timestamp, where the origin node represents a patient and the destiny is an exam name. Also, each timestamp in this network was converted as a single day. Figure 2 shows the resulting network after the sampling, and Table II summarizes the information for each healthcare institution after the guided sampling (compare to Table I). To calculate the number of communities, I-CovidVis uses the weighted Louvain method, which is based on modularity optimization and is one of the most popular and recommended methods [4].

I-CovidVis follows the cornerstone approach of “*Overview first, zoom, details on demand*” [21]. We first analyzed the uni-

Facil.	Edges	Patients	Exams	Days	Com.
HF1	5,425	4,160	14	285	14
HF2	14,715	13,905	19	274	18
HF3	4,592	1,236	41	168	9

fied data from the three healthcare institutions to identify intersections and differences, and to validate the interoperability capabilities of our approach. Then we analyzed the data from each institution to understand each institution’s characteristics related to the global perspective. Finally, we performed a local analysis, tracing clinical paths, as exemplified in Figure 4 for patients from HF3 who had the outcome “death”.

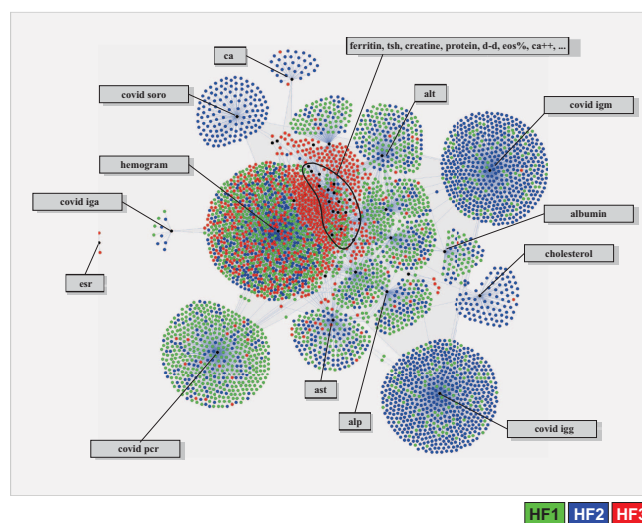


Fig. 2. Global perspective for the FAPESP database with the network balanced for 2,000 edges per healthcare institution. Each color represents a different institution.

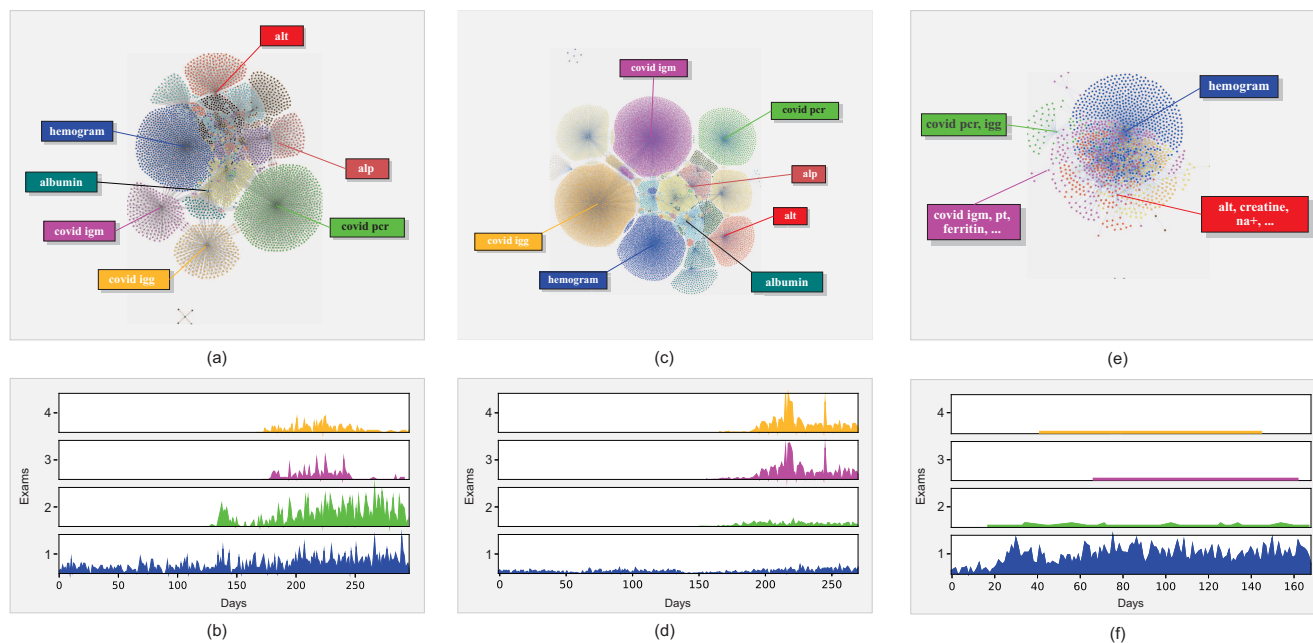


Fig. 3. Visual analytics of HF1 (a-b), HF2 (c-d), HF3 (e-f) with node-link diagrams (a-c-e) and line chart (b-d-f) layouts.

tested, there is a group of exams (ferritin, aptt, tsh, creatine, protein, ggt...) closer to hemogram, indicating that these exams are tested together, mainly in HF3.

Figure 3(a-b) shows data from HF1, according to two perspectives: node-link diagram (a) and line charts (b), which represent the number of exams performed by timestamp. The colors are set by the community detection algorithm and the nodes are positioned by the force-directed layout. Although the two techniques are based on the same principle, not always generate compatible results, such as the same color for equivalent groups. In the case of HF1, the communities represent exactly the number of exams, i.e., each community contains a single exam (see the equivalence of Exams and Communities in Table II). This pattern eases analyzing the node-link diagram since it allows seeing patients strongly related to each exam. It is important to notice that, although a patient is related to a single exam, it does not imply that the patient has just a single exam, but most probably the patient just performed an exam more often than the others. To exemplify a temporal perspective analysis, Fig. 3(b,d,f) compares four exams among the three institutions: Hemogram (1), Covid PCR (2), Covid IgM (3), and Covid IgG (4).

The peripheral nodes in Figure 3(a) show well-defined communities of different sizes, with Hemogram and Covid PCR representing the most performed exams in HF1. The central region of the layout exhibits a denser area, revealing exams that had fewer tests and patients that had several tests but no predominant one. Another important insight from this data is the line chart, shown in Figure 3(b). It is a coordinated view with the node-link diagram, where each color represents the same exams. I-CovidVis show that this healthcare institution

started providing exam results long before the first covid PCR tests, allowing tracing the history of the analyses. By analyzing the temporal perspective of IgG and IgM (Exams 3 and 4), it can be seen that around day 250 there is a decrease in the number of these tests in the healthcare institution and an increase of PCR tests. This pattern indicates that due to PCR availability the treatment protocols changed, which resulted in the reduction of IgG and IgM and the increase of PCR test.

Figure 3(c-d) evaluates HF2, which has a structure similar to HF1. Both have the detected communities with almost the same number of exam types (respectively 18 against 19). As there is a strong relationship among communities and exams, it is noticeable that overall the most tested exams are covid IgG and IgM, followed by hemogram and PCR. Furthermore, the institution also provided the exams taken before the first covid tests. Although IgG and IgM were overall frequently tested, it is also possible to see a decrease of testing after day 225, similar to the pattern detected in HF1. This behavior is consistent with protocol changes over the first months of covid tests, in which PCR became the most suitable option for most institutions. I-CovidVis also shows some exams positioned closer to each other, such as alt, alp, and albumin, indicating that there is a group of patients consistently tested for those related exams.

Figure 3(e-f) shows results from HF3, which has a very different structure as compared to HF1 and HF2. Here, the relationships among communities and exams are small, with only 9 communities for 41 exams. This behavior can be understood by the high proportion of exams taken per patient (see Table I), i.e., the patients are tested for a wider variety of exams when compared to HF1 and HF2. Moreover, in our

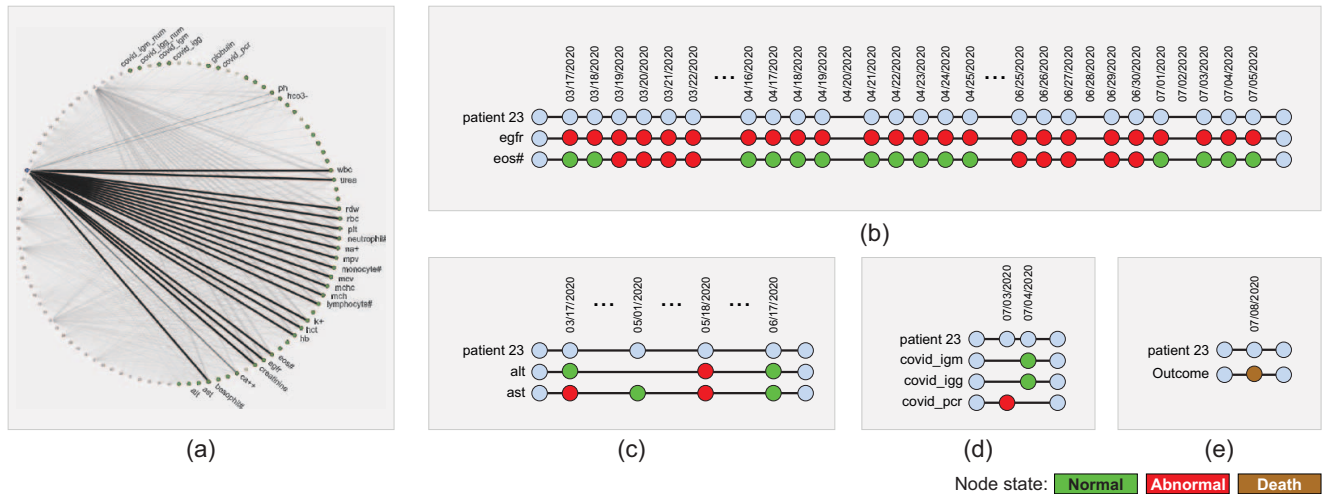


Fig. 4. Clinical path for HF3 highlighting the patient 23 activity over time.

sampled size, most of the communities tested for more than one exam. Distinctly from HF1 and HF2, the distribution of the communities can represent patients with related symptoms, and not to groups of different exams. From a temporal perspective, it can be seen that hemogram was the most tested overall and there are also days showing a decrease of temporal activity (e.g. between days 40 and 60).

Although a global perspective is important to understand the data structure or how it evolves, such as shown in Fig. 3, it is also important to understand the local patterns. Thus, more specific analyses can be done for small groups of subjects, which can be achieved at the visualization interaction steps, which our system provides. Figure 4 illustrates a case study for HF3, containing only 44 patients that had death as the outcome. To understand how a patient came to an outcome, I-CovidVis yield the patient's clinical path, which shows the exams tested, what were the results, and at what times each patient had the exam. It allows to interactively coordinate and interact between the node-link diagram and MSV views, to select groups of exams related to a patient, and analyze the exam results that lead to an outcome.

Figure 4(a) illustrates a circular node layout in the node-link diagram, where thicker and darker edges represent exams that were tested more often. Patient id 23 was selected for this figure, which shows all exams that he/she performed. Figure 4(b-e) represents the MSV layout with selected exams performed by patient 23, only showing the node-activity and hiding the edges. I-CovidVis indicates an exam with a result as green (normal) or red (abnormal). Figure 4(b) shows the relationship between egfr and eos# exams for patient 23. Notice that all tests of egfr were abnormal, while the eos# changed over time, and that those two analytes were tested over several consecutive days. The egfr results demonstrate that the patient has chronic kidney disease. Figure 4(d) shows that this patient was tested for covid just a few days before the outcome, indicating that hospitalization and in-hospital

contamination have occurred.

Figures 5(a-d) show a second case study using the clinical path, now for patient 31. The first analysis (Fig. 5(a)) evaluated his/her activity in relation to analytes aptt, ast, alp, and alt. The first test for those analytes indicated normality. However, during two consecutive days, aptt presented abnormal results. Four days after, the patient was tested for all those four analytes and all results were normal. Furthermore, the figures show that this patient was tested again over the next months for these analytes, having mixed results. Among other tests, it is possible to see covid tests in two months, first indicating contagions (PCR abnormal) and then indicating the recuperation (PCR normal and IgM and IgG abnormal). Nonetheless, although there was a wide variety of tests often indicating normality, the patient reached death possibly due to other comorbidities (Fig. 5(d)).

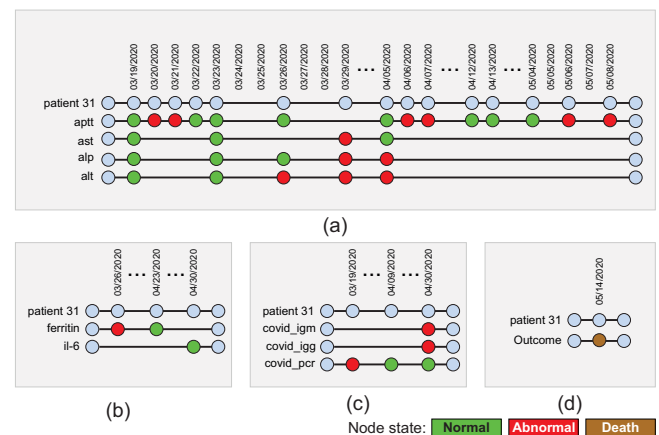


Fig. 5. Clinical path for HF3 highlighting the patient 31 activity over time.

V. DISCUSSION AND LIMITATIONS

The analyzed database, as others from real applications, lacks a detailed description of the variables, laboratory methods involved, acquisition and calibration protocols, and measurement errors. This is aggravated by the use of simplified internal vocabularies, varying notations, and optional observations written in free text, which make it difficult to treat all the data at once in a consistent way. Also, as the database contains different institutions, some of them could have released more exams per patient in their repository construction, which may not represent its real scenario. Another shortcoming is how to interpret different laboratory methods to measure the same analyte, so they can represent the same measured quantity (or not), which requires careful discussion between the biomedical specialists and what level of accuracy is required for subsequent diagnostic hypotheses. Another limitation is the lack of previous patient history, medications in use, and follow-up outcomes, which limits both the causal investigations and the application of personalized medicine.

There are some aspects of the visualization perspective that require further attention, such as the number of colors, support for color blindness, parameterization of the techniques, visual scalability, and so on. For instance, several similar colors to represent different communities can be a problem due to the eyes' limitation of differentiating the colors. In this case, redundant coding can help to improve the visual analyses. Furthermore, while the focus was not on visual and computational scalability, the visualizations worked well for more than ten thousand nodes and edges (HF2) with all processes executed in linear time. The guided sample method employed got samples from every laboratory, maintaining a proportionality to the total number of tests. However, this strategy can lead to inconsistencies, as less frequent tests could nevertheless influence the visualization. On the other hand, maintaining all tests would make it hard to see the whole picture due to visual clutter, thus making available varying sampling options can make sense.

VI. CONCLUSION

Visual data analysis is an important tool to understand complex systems, relations, and to find unexpected patterns. In this paper, we introduce I-CovidVis, a visual analytics tool focused on analyzing interoperable healthcare databases using graphs, aiming at helping understand an institution structure, cases evolution, relationships among exams, patients, and its clinical path. In our case study, we analyzed data from three different healthcare institutions from a global perspective using an aggregated structure, down to compare similar characteristics and to visualizing each one separately, enabling the understanding of individual behavior and evolution path. For local analysis, I-CovidVis provides resources to visualize clinical paths and to coordinating and interacting with views between the graph structure and a temporal path, which helps the health professionals to analyze exam results and understanding which steps lead to a patient outcome. Due to space limitations, we did not show complete clinical paths for

a patient. However, I-CovidVis allows the specialist to interact with the visualizations, focusing on specific parts. For future developments, we intend to adapt our system for standard data schemas, like OMOP-CDM, also considering varying behaviors and techniques to select similar patients to compare their clinical paths. Furthermore, we intend to perform user evaluation with specialists to improve the tool usability.

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